

Practice Quiz: Agents

Name: __ Date: __

Part A: Multiple Choice

1. In Active Inference, what makes a system an agent? A) It has a physical body B) It possesses a generative model — an internal model that predicts sensory observations C) It can move D) It has a brain
2. A generative model includes: A) Only beliefs about the current state of the world B) Beliefs about hidden states, how they cause observations, and the agent's preferences C) Only preferences about what the agent wants D) A complete and perfect map of reality
3. Prediction error is: A) The cost of making a prediction B) The difference between what the agent predicted and what it actually observed C) A type of logical fallacy D) Only relevant when the agent makes a mistake
4. An agent can minimize prediction error by: A) Only changing its beliefs (perception) B) Only changing the world (action) C) Either changing its beliefs (perception) or changing the world (action) D) Ignoring the prediction error
5. The self-model is: A) A mirror image of the agent B) The agent's internal model of its own body, position, and mental states C) Only found in humans D) The same as the generative model of the environment
6. Proprioception is: A) The sense of smell B) The ability to see in the dark C) Your body's sense of its own position and movement D) The sense of touch
7. Which of the following is the best example of an agent updating beliefs rather than acting? A) Opening an umbrella when it starts raining B) Realizing it's raining after hearing sounds on the roof C) Running indoors to avoid the rain D) Checking the weather forecast before going outside

Part B: Short Answer

1. A student expects to do well on an exam but receives a poor grade. Using Active Inference terminology, explain: What was the prediction? What was the prediction error? Give two different ways the student could minimize this prediction error (one through perception, one through action).
2. Is a plant an agent? Using the criteria from this module (generative model, prediction, prediction error, action), argue for or against. Consider: a sunflower turning toward the sun, or a plant's roots growing toward water.
3. Explain the difference between perception and action in Active Inference. Why are both needed? What would happen if an agent could only perceive but never act, or only act but never perceive?